



Exploratory Data Analysis
Project



The Case

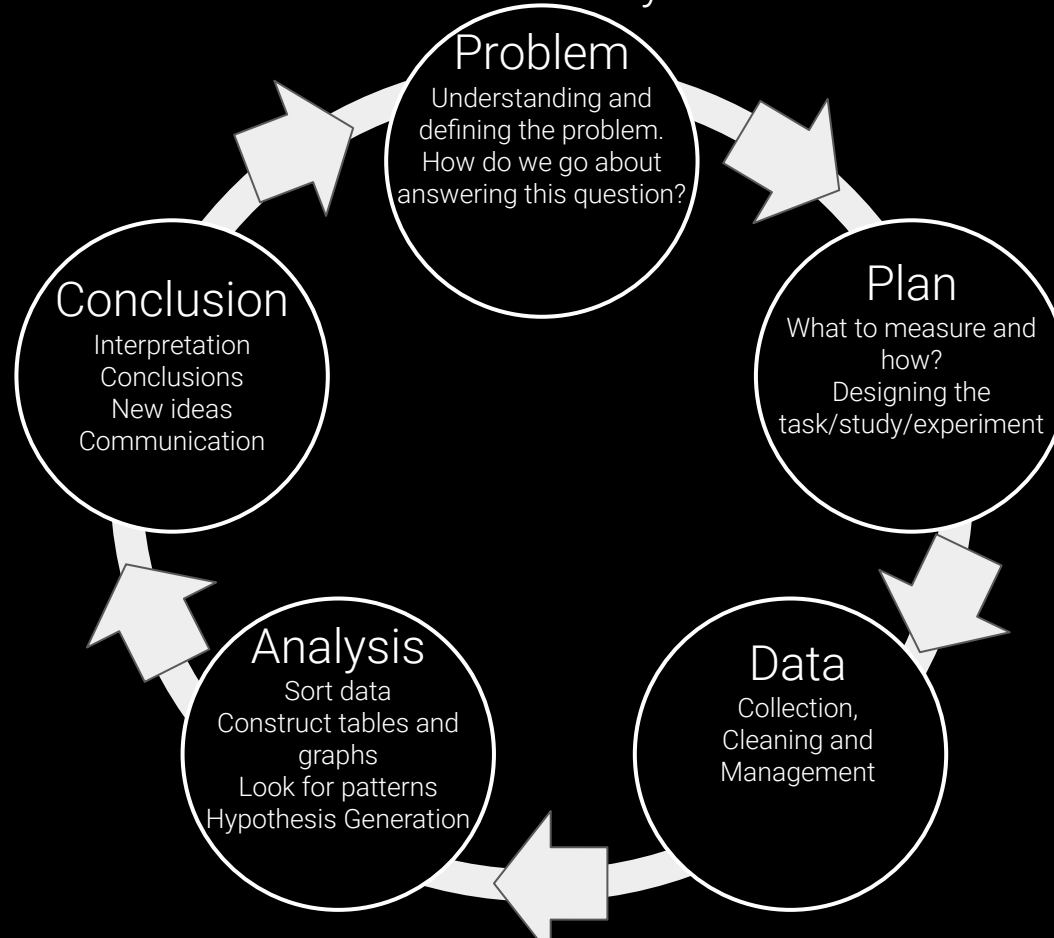
You will be given a dataset ``order.csv`` containing orders from an e-commerce company.

Axel and Philip are stakeholders in the company and we have hired you as Data Analyst Consultants to find and communicate business intelligence insights in the data.

You will do the case in pairs of two.

Working Methodology?

The PPDAC-cycle





Data Storytelling

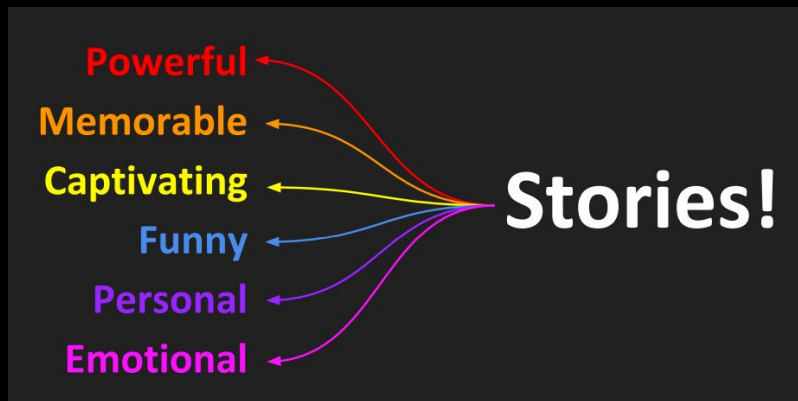
- Data is often seen as:





Data Storytelling

- While stories can be so much more!





Setup, Conflict, Resolution

A Simple Framework For Storytelling With Data

Setup

This is where you introduce the context of the data. It involves setting the scene for your story by explaining what the graph represents, including the type of data, the time frame, and any relevant background information. This step is crucial for helping your audience understand the importance of the data and what to pay attention to in the graph.

Conflict

In this part, you highlight the challenges, trends, or anomalies presented by the data. The conflict is essentially the "plot" of your data story. It's where you point out any unexpected findings, significant changes over time, or comparisons between different data sets or categories. This step is about engaging your audience's curiosity and emotions, making them invested in finding out why these patterns exist and what their implications might be.

Resolution

Finally, you provide a conclusion or insights based on the data. This is where you interpret the data, offer explanations for the patterns observed, and possibly suggest actions based on these insights. The resolution should wrap up the story by addressing the "conflict" presented earlier and helping your audience understand what can be learned or what steps might be taken next.



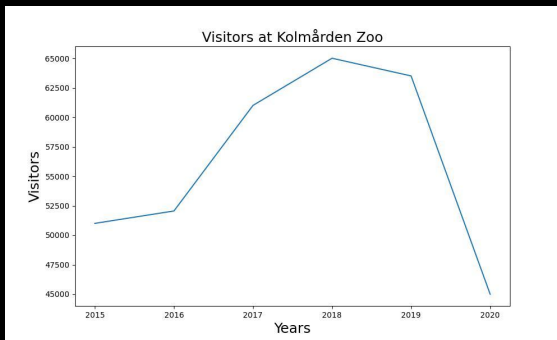
Setup, Conflict, Resolution

An example

Setup: Begin by setting the scene for your audience

Conflict: Introduce the conflict by highlighting the trends and anomalies

Resolution: Provide a resolution with insights and conclusions



Setup

"Today, we're looking at the annual visitor numbers to Greenwood National Park from 2015 to 2020. This graph not only shows us the sheer volume of people attracted to the park's natural beauty but also helps us understand trends in outdoor recreational activities over these years."

Conflict

"As you can see, from 2015 to 2018, visitor numbers increased steadily, reflecting growing interest in outdoor activities. However, there's a noticeable dip in 2019, followed by a significant drop in 2020. This deviation from the trend raises questions about what could have caused such a change. Was it due to external factors like wildfires or park renovations, or perhaps the impact of global events like the COVID-19 pandemic?"

Resolution

"Upon further investigation, we find that the decrease in 2019 was due to several months of park closure for infrastructure upgrades. The dramatic fall in 2020, however, aligns with the onset of the COVID-19 pandemic, leading to travel restrictions and nationwide lockdowns. This insight not only explains the visitor trends but also highlights the park's resilience and the public's fluctuating interest in outdoor activities based on external circumstances. It suggests that while the park faced short-term challenges, the overall interest in natural spaces remains strong, indicating potential for recovery as situations normalize."



Project Requirements

Find interesting insights in the data that provides value to the stakeholders

Visualize and communicate the insights in a clear manner

Follow to the PPDAC cycle when analyzing and visualizing the data

When you find an insight in a plot you write down key words that follows the setup-conflict-resolution framework.



What is Value in Business

Business needs to make money to survive, that means that revenue needs to be higher than the costs. If the business' revenue is higher than their costs, they make a profit. Profit is the shit. Profit can be increased by increased revenue, decreased costs or/and increased prices.

If you want to advocate an action a business should take, you have to show how that action will be net positive in terms of profit. If you can motivate how this will make money for the stakeholders, you will get your way.

In business, we like benchmarks. Showing that our revenue is many many million dollars doesn't mean jack shit unless it is put in relation to something else. For instance, we relate revenue to cost and calculate a profit margin.

Customers are important and the most profitable thing is to recurring customers, they are the best. It is cheaper to keep a customer than to acquire a new one. By understanding the behavior of customers, we can target customer in a way that is appealing to them and increase sales. It is always cheaper to keep an existing customer than to acquire a new one.